

# Mid-band public multimedia kiosks

J A Totty

*Multimedia kiosks are becoming widespread as companies in the retailing and service sectors search for new ways to do business while reducing the cost of their high street presence. The paper gives the rationale behind these products, describes current options for multimedia service delivery to public sites, and highlights the developments necessary to ensure that kiosks can 'close the sale'.*

## 1. Introduction

Multimedia kiosks are public-access screen-based systems, providing the general public with temporary access to information, sales or entertainment services, the purpose of the kiosk being to secure from the user a commitment to purchase goods or services, and to provide instant fulfilment where possible.

Kiosks are appearing in retail (including retail finance) outlets and travel sites, but the installed base has remained relatively low, reflecting uncertainty on the part of site owners, often resulting from a lack of established kiosk technology, networking and content management on the supply side. Happily, the position seems to be changing, as there is a growing desire on the part of both content providers and major IT companies to enter kiosk markets as a precursor to the exploitation of larger business and domestic markets.

## 2. Sites and services

Kiosk installations can currently be found in a variety of sites, but from a commercial perspective there are only two site types. The first is the **private** site, where a kiosk sells the site owner's products or services (and little else). Examples of private sites would be banks, building societies, tied car showrooms, shops and supermarkets. Two or more private site owners may of course enter into cross-selling arrangements. The second type is the **public** site, in which the site owner will allow a broader range of products and services to be sold through the kiosk, examples being airport lounges, bus and coach stations, shopping malls, local authority sites, town centre pedestrian areas, railway stations and motorway service areas. Kiosks in the latter group are often described as 'multiservice' kiosks.

Typical services offered range from simple (passive) advertising, through catalogue browsing (e.g. browsing information on financial services, holiday destinations or products for sale) to specific searching (e.g. what flights are available at this time). The kiosk can provide a telephone, videophone or data link to a service-specific sales desk for

the handling of enquiries, and can provide the user with hard copy of relevant information (e.g. a map showing the location of a hotel). A 'top of the range' kiosk (see Fig 1) can confirm an order or reservation, take payment for the



Fig 1 Demonstrator multiservice kiosk at BT Laboratories.

selected product or service, and provide a receipt for the transaction. Such kiosks are described as 'full function' kiosks. For the purchase of printed information, travel or venue tickets, or for the issue of vouchers, instant fulfilment is possible. Where consumer products or very high value services are being bought, fulfilment requires collection or delivery by external means.

Non-retail services can be envisaged, particularly where there is a need to make central government or local authority services available to the general public. Electronic benefits payment and local authority information are valid future offerings in this context, as is televoting.

A further set of potential services rotates around games and gambling, and the success of the UK National Lottery suggests some potential in this area. Kiosks are already installed in some betting shops.

### 3. Kiosk configurations

Since the major drivers for kiosk evolution are to be found in retailing, kiosk technology has reflected a desire to emulate traditional selling methods. Specifically, kiosk developers, driven by vendor needs, have attempted to replace, one by one, the four recognisable stages in the selling process, which are:

- to capture the buyer's attention and retain attention during advertising, appealing to both reason and emotion by positioning the product in a lifestyle context, linking the vendor's product and brand with values that are shared by the buyer,
- to provide detailed information, emphasise benefits, offer options,
- to enter into dialogue with the buyer (if necessary), understand the buyer's circumstances, and eliminate objections to sale,
- close the sale, secure payment or commitment to buy, initiate fulfilment process, and congratulate the buyer.

Kiosk terminals can be grouped into four types, reflecting their function, namely POI terminals, browsers, option selectors, and full-function kiosks.

POI (point of information) terminals give information in a predefined order only.

Browsers permit free roaming by the user but do not offer any user interactivity beyond navigation, achieving the first two stages of the selling process, but requiring human intervention to continue the sale.

Option selectors allow users to assemble a product or service package according to their liking, and to register an interest by inserting a closed user group card such as a retail store card, or having a product description voucher printed which can be taken to a sales desk for manual processing,

achieving the first two stages and some elements of the third stage in the selling process.

Full-function kiosks provide a sales desk channel for the handling of customer enquiries (voice or videotelephony), and a means of handling financial transactions and receipt printing, achieving all stages of the selling process.

In all cases, keyboard interaction is generally eschewed in favour of touchscreen, the combination of active pictorial elements (buttons or icons), and touchscreen being particularly well suited to applications intended for naive users.

It is worth noting that all four types of kiosk have traditionally stored their multimedia content locally, content being pre-recorded on hard disk or CD-ROM residing within the kiosk cabinet. As a result, there has been little interest in large-scale networking of kiosks, except for the carriage of transaction traffic (established in the automatic teller machine service) and voice calls stimulated by the fixed content. However, the emergence of suitable networked multimedia authoring and run-time software signals an end to the traditional total dependence on local content. Figure 2 charts the evolution of multimedia kiosk technology to date, and shows the emergence of remotely sourced content delivered via network.

In the stand-alone configuration (currently the most common arrangement to be found in live installations) content is held within the kiosk, and no networking is invoked. Stand-alone kiosks are suitable for basic browser applications, particularly where the content has a low churn rate.

The configuration can be extended to provide a low bit rate data line or voice telephone line to a service provider back office, allowing sales or service enquiries to be handled, or simple data transfers such as account balance check or goods in stock check. The configuration is particularly suitable for option selectors in private sites, and echoes current trends in catalogue shopping and automatic teller machine (cashpoint) system design. For small amounts of content with low churn, it is possible to edit the fixed content from the remote helpdesk site by file transfer.

The on-line multimedia configuration extends the capability of the kiosk still further by holding the multimedia content at a remote server, allowing for high content churn rates, and offering access to a potentially large selection of services, but requiring a trade-off between audio-visual richness of the content and network economics. It is particularly suitable for multiservice kiosks in public sites, or for private sites with significant service variety and churn. Enquiry and transaction lines can be provided, as in the option selector case.

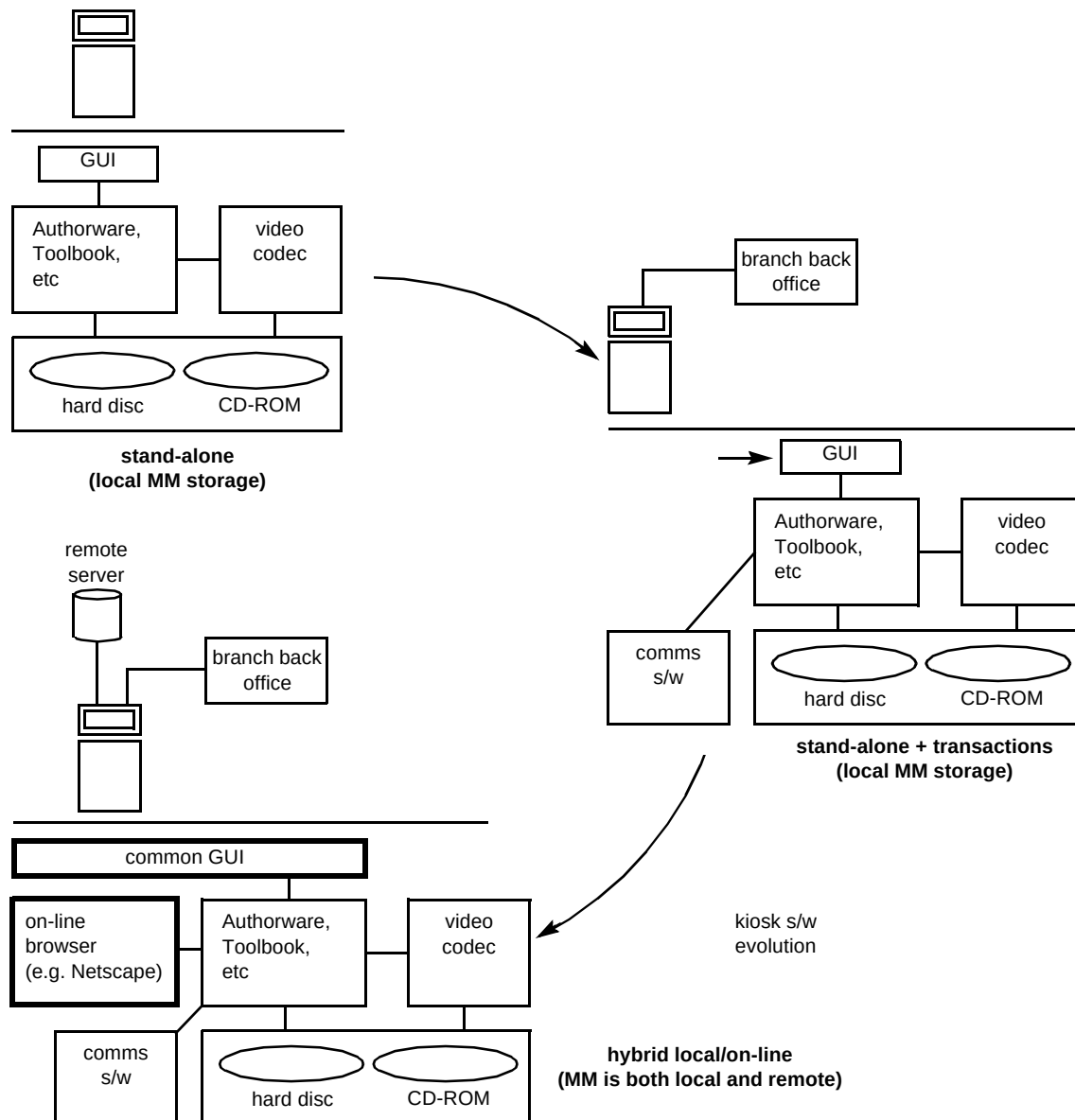


Fig 2 Evolution of kiosk multimedia configurations.

#### 4. On-line architecture, mid-band solution

It is possible to design scenarios around any one of a number of CPE/network combinations, including TV or PC with copper, fibre, or terrestrial or satellite radio. For BT's kiosk activities in the short term, however, a combination of PC and ISDN networking gives the optimum solution:

- offering excellent geographical coverage (existing network base),

- using ubiquitous terminal hardware (the PC) and peripherals,
- offering synergy with home and office PC information services market,
- using existing multimedia formats (e.g. Worldwide Web formats),
- using available (and evolving) service creation software,
- requiring relatively simple server environment,

- supporting telephony, videophony and transaction handling,
- offering service flexibility, anticipating future regulatory frameworks.

The major disadvantage is the lack of online richness resulting from a constrained bandwidth of  $2 \times 64$  kbit/s. Nevertheless, suitable on-line multimedia service creation and run-time software packages are available, suppliers being lured by the growth of Internet Worldwide Web (WWW) use, and major suppliers are forming alliances to ensure continued development in this field.

Figure 3 shows possible relationships between a range of kiosk types and possible multimedia activities in other mass-market areas, and suggests that there may be a spectrum of kiosk applications with high service richness at one end, and high interactivity at the other.

Figure 3 also shows that services requiring high levels of factual content and interactivity for 'getting things done' are clustered around PSTN and ISDN as preferred channels to the business community, typically Internet WWW services and information services brokered by national and international on-line 'supermarkets'. Entertainment services are shown mated with broadband technology as a delivery mechanism.

## 5. Multimedia content transport

In the stand-alone kiosk, content in the form of text, still pictures (e.g. JPEG encoded) and motion video (e.g. MPEG encoded) is stored on the kiosk PC hard disk or CD-ROM. The content is bound together by hyperlinks which are added during the service creation process and which control the sequence of activities when a particular screen icon is touched. Multimedia authoring software packages such as Authorware or Toolbook are supplied with built-in run-time functionality, the same software package being used both to create the navigation code (the hyperlink 'glue') and to play back the finished product. To drive a printer or card reader, the navigation code points to a process file which contains the necessary instructions to drive a particular peripheral.

In the case of on-line multimedia, two methods are employed for transferring content from the remote server to the kiosk, namely bulk file transfer (block downloading of entire service scenarios) and on-line browsing (piecemeal downloading of one or two pages at a time to PC RAM-cache).

Although **bulk file transfer** is practised by at least one kiosk supplier, the technique requires broadband networking to update content within reasonable times,

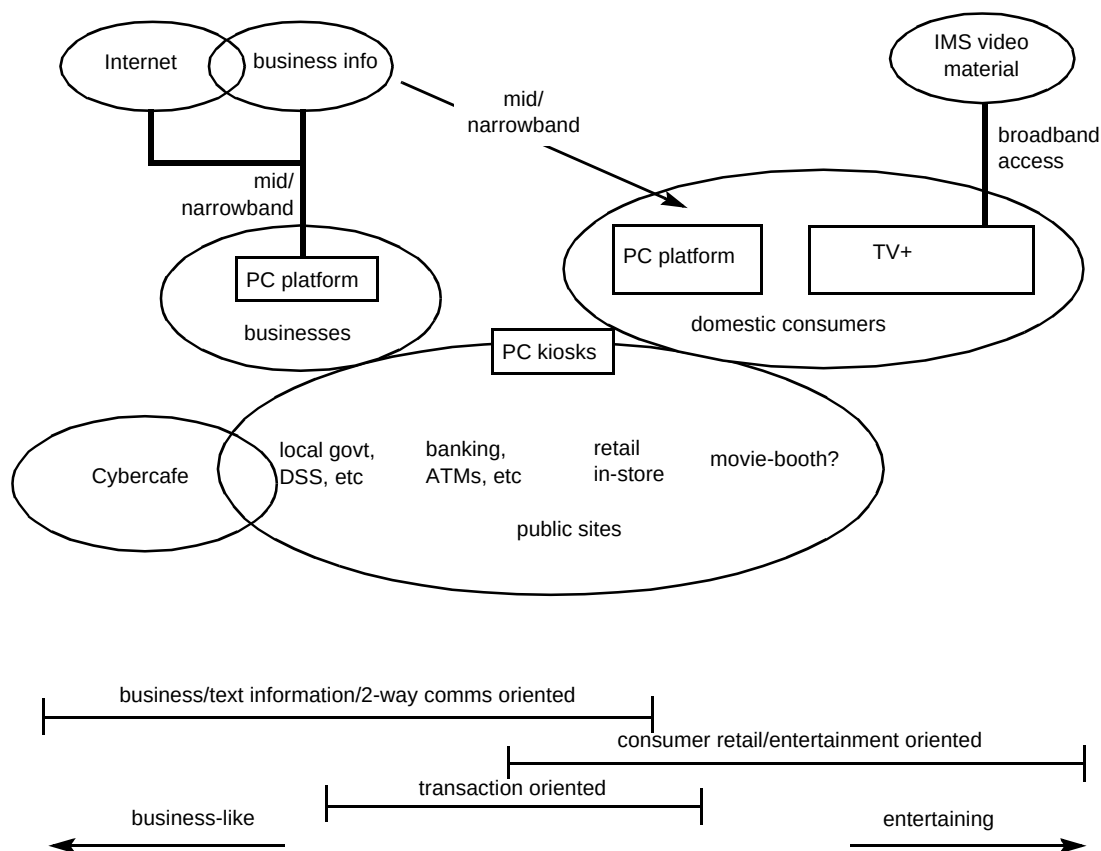


Fig 3 Kiosk applications in relation to business and domestic multimedia.

although it does allow video-rich scenarios to be downloaded, and makes kiosk configuration management relatively straightforward. Block downloading via ISDN2 is time consuming (a 3 minute MPEG video scenario could take up to 40 minutes to download) and presents scaling problems with growing kiosk populations.

**On-line browsing** is favoured by a number of kiosk manufacturers since this is the mode of operation of the growing Internet WWW service, and browser software such as Netscape or Mosaic is readily available, allowing access to hyperlinked content usually in the form of text and still pictures, the content data being formatted in hypertext mark-up language (HTML) and loaded on to a relatively simple UNIX or similar server. One interesting feature of on-line browser software is the ability to compile a page (such as a menu page) 'on the fly' by assembling the necessary hyperlinked visual images (e.g. icons) on to a page grid in real time.

On-line browser software packages are not able to handle real time video content at this time. It should be noted that it is not possible to deliver full-screen MPEG video in real time over ISDN2 at all, although H.261 video (lower motion fidelity) can be delivered using the cards and software from PC videoconferencing systems. On-line browser software suppliers plan to include MPEG video

capability in the near future. However, the position for ISDN2 networking is that full-motion video sequences, where required, have to exist as locally held content, which implies that site visits will be an element of service creation and churn management, unless the sequences can be so short as to permit scalable downloading to kiosks.

A number of kiosk manufacturers have developed hybrid kiosk software, which fuses together the fixed content tools (e.g. Authorware) and on-line browser tools (e.g. Netscape) under a seamless graphical user interface. For private sites with only one or two main services, this is an ideal solution. For multiservice kiosks where churn is high, the options are clear — content must either be created for browsing only, or else the logistics associated with holding video content locally have to be planned in advance and costed into the service creation process.

## 6. Richness versus interactivity

This picture of on-line and supplementary fixed content is explored in Fig 4, which positions a number of multimedia services in terms of the content richness and level of interactivity offered. Richness in on-line services is a function of available bandwidth, but can be boosted by the local addition of supplementary (fixed) content. Interactivity is a

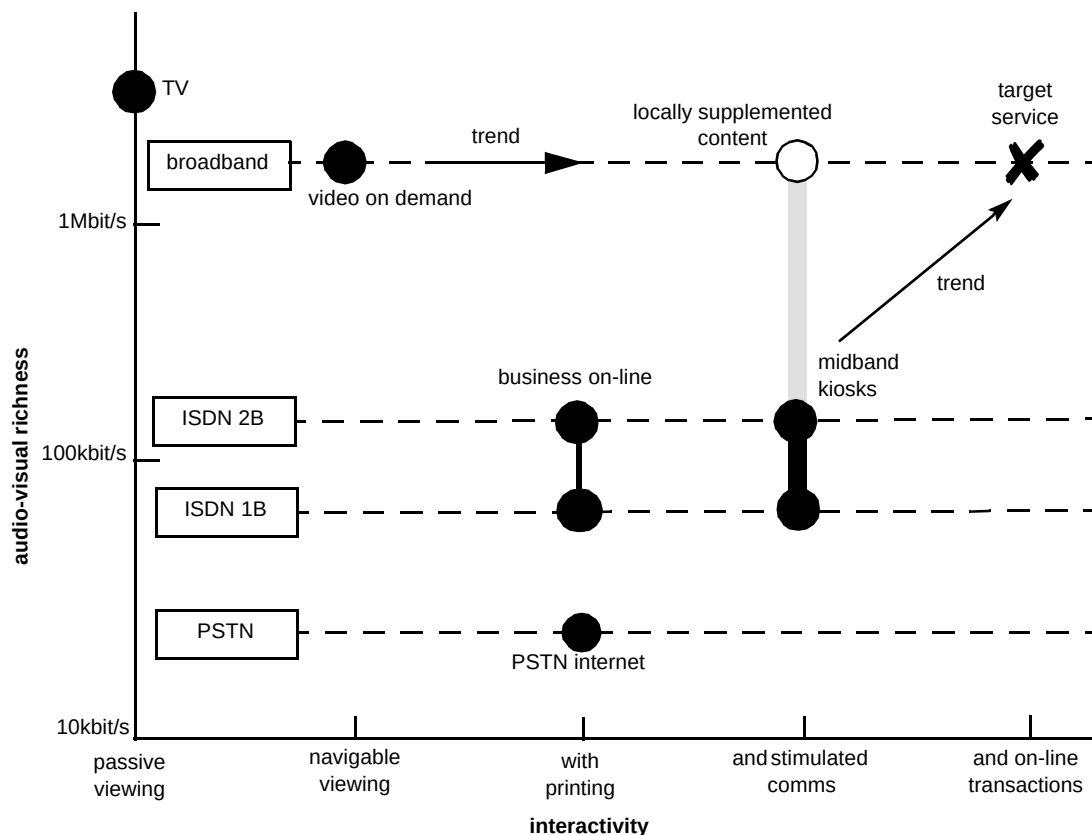


Fig 4 Richness versus interactivity for a selection of services.

function of the kiosk content navigation design, the peripheral equipment level and the provision of communications links and transaction handling systems.

Since the *raison d'être* of kiosk services is fast turn-round, the combination of high interactivity and intermediate richness is an appropriate solution. In fact, for most kiosk applications, video sequences lasting more than 2 minutes will have a counter-productive effect. Crisp, imaginatively designed pages and good audio can be just as effective as video, despite the power of full-screen video to reach users at an emotional level or to explain abstract concepts. The key feature of any useful kiosk service, then, is interactivity.

## 7. Kiosk interactivity

Kiosk interactivity includes the following three components:

- the degree of user control over the content displayed (navigation),
- the terminal peripheral equipment level (printers, etc),
- provision of networked dialogue and financial transaction handling.

The degree of user control over the content displayed varies considerably across multimedia applications. For video-rich applications, a tape-transport analogue is often used, offering 'play', 'stop', 'review' functions. For the reasons already outlined, this mode of control has little value in kiosk environments, and the preferred mode of operation is via hyper-linked icons (pictorial or labelled button menu items) to successive pages or screens, allowing the user to quickly funnel into the product or service of interest. The need for quick turn-round of business suggests that menu trees on kiosks should be shallow (no more than five layers deep) and self-explanatory.

Terminal peripheral equipment levels vary considerably across terminals according to the target application, but to achieve the level of interactivity required on a full function terminal, plain paper printing, voucher printing, receipt printing (local or with on-line validation), cash (coin) validation, magnetic stripe card reading and smart card reading are all necessary in some combination. Fortunately, most of these peripherals are either standard office equipment products or used in point of sale products, and the terminal design problem becomes one of integration and packaging.

The real power of the kiosk is in its ability to stimulate calls to sales helpdesks or advertisers, and to handle

financial transactions. Both require networking, but local line contention may occur if a browsing instruction is made at the kiosk while a sales desk call is in progress. A simple solution is to provide the kiosk with a separate sales desk/transaction channel in the short term, allowing voice calls to be set up directly by touching an advertising icon on screen or by touching the 'sales helpdesk' icon within a specific service. The sales helpdesk channel can be extended to link the kiosk to the sales back office data system to permit price checking or transaction handling.

## 8. Payment methods

Any kiosk with a line to a remote sales desk can provide financial transaction handling by the (insecure) method of having the user quote their credit card number over the voice channel. For the purchase of goods, fulfilment requires that goods are delivered to the credit card owner's registered address, which goes some way to overcoming fraud problems. The use of magnetic stripe cards in (most) unattended terminals is currently forbidden by card issuers. However, the introduction of smart card payment technology should permit direct transfers between unattended kiosks and sales desk back-office systems in the near future, smart cards being relatively resistant to cloning fraud. During the transition phase between magnetic stripe and smart card use, hybrid card readers will be required. Such readers are already becoming available and will be procured in volume for payphone products.

At this time, two types of smart card are of interest, namely the token card and the electronic purse card. The token card is a prepayment card which allows units of credit to be spent on predetermined services only. Examples are the prepaid phonecard and parking cards. When the credit is spent, the card is either thrown away or topped up by a recognised dealer. The electronic purse card, e.g. Mondex, is in essence electronic money, which can be spent or accumulated in any way (including from person to person) provided the funds transfer electronics exist. During a transaction, the digital equivalent of pounds and pence is carried over the network, and considerable effort has been expended on encryption to make fraud (almost) impossible. The electronic purse card offers the prospect of secure teleshopping without the need for service subscription and is an ideal future payment mechanism for kiosk services.

## 9. Kiosk management

For private kiosk systems running on LANs and accessing a local server, system management is a modest extension of standard office automation practice, the added complexity being the need for content creation and installation. For large, geographically distributed systems working

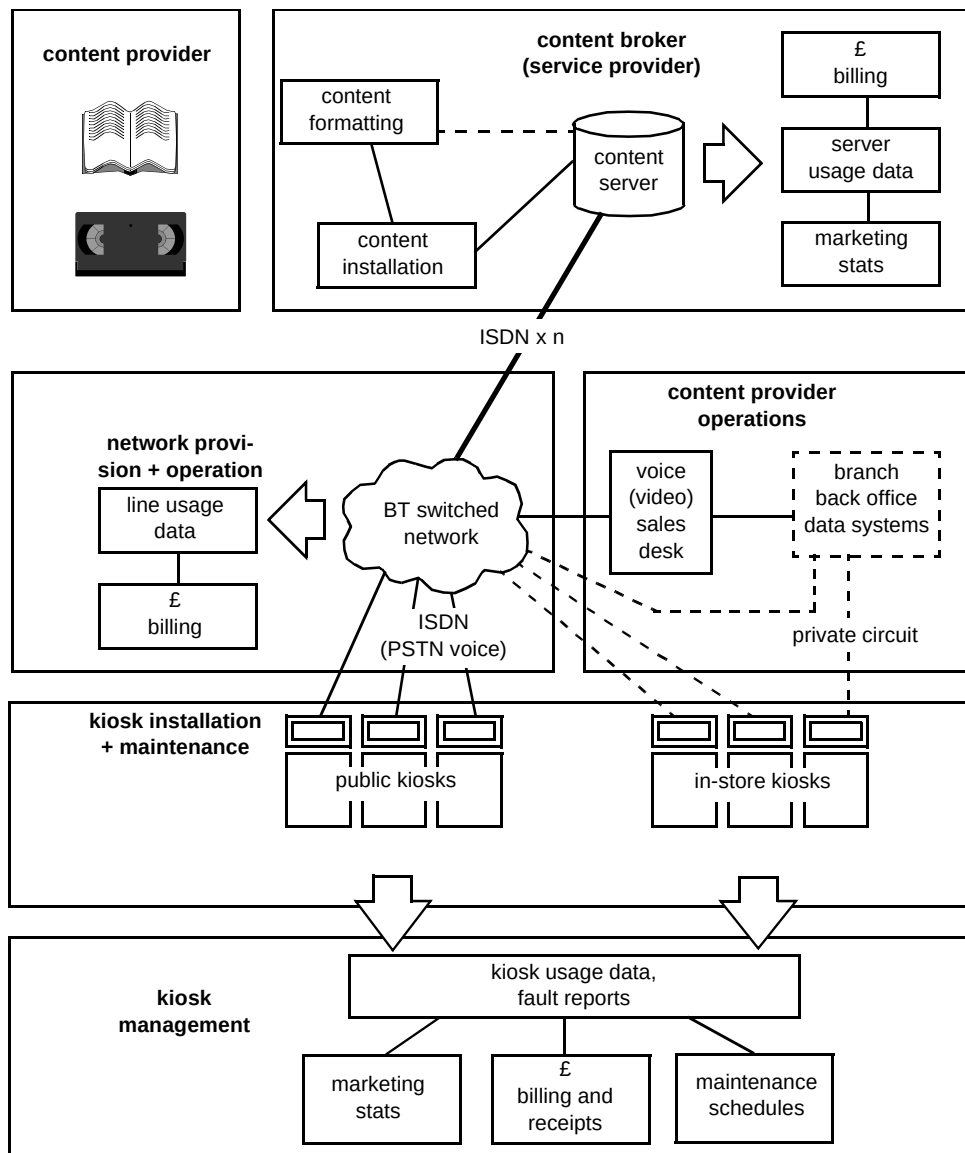


Fig 5 Networked kiosk architecture and core service management system.

over the public network, service management requires significant effort. While undoubtedly adding cost to the system, service management should also be seen to have revenue potential.

Figure 5 shows a potential national architecture for a large kiosk system, and illustrates the need for management processes and systems in the server, network, and kiosk terminal domains.

Surrounding the core service management systems, and not shown in Fig 5, are the traditional service operational support activities including provision, maintenance and repair, helpdesk and content management.

Kiosk and content usage data can be translated into marketing statistics which allow content providers, service provider (broker) and Kiosk operator to assess how well

services perform in certain site types or geographical areas. The usage data also provides the basis for billing and reconciliation activities. From the terminal operator's point of view, kiosk alarms and fault reports need to be processed quickly, linking in to existing automated maintenance systems where possible.

## 10. Futures

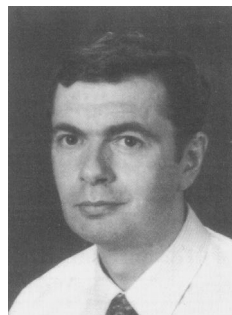
The future of kiosks is probably best suggested by assuming that technologies will evolve to meet pressures for more richness and interactivity across growing mass IT/communications markets.

A small increase in richness may arise as browser software evolves in the short term. In the longer term, richness may be met over restricted bandwidth channels by the transmission of a 3-D image to the kiosk and its

subsequent animation on screen, but a more likely scenario is that kiosk architectures will yield to broadband networking growth, allowing more service 'presence' at the kiosk.

Opportunities may arise in evolving electronic payment fields, particularly smart cards, eliminating the need for user subscription, and allowing spontaneous interaction. Cards holding user identification data will allow individual spending patterns to be tracked, leading to precise targeting of new product or service offerings.

The scope of kiosk services will grow to encompass service customisation, electronic benefits transfer, televoting, gambling, and arcade style games. In the longer term, content granularity will change as individuals assert their right to become content providers.



Andrew Totty is a solution designer within the design and build directorate at BT Laboratories, and lead designer for multimedia kiosk activities on behalf of BT Core Multimedia Service Provision and BT Payphones.

He graduated from Leicester University with a first degree in electronics and a masters degree with distinction in industrial design. He was involved in weapons platform design for the Admiralty and instrumentation system design for the Health and Safety Executive before joining Marconi Avionics (Rochester) in 1980 as an aerospace design engineer. In

1983 he moved to consumer product consultancy practice in Oxford, covering capital and consumer products for major UK clients, before joining BT in 1987, since which time he has been involved in high level product and service definition and has managed 'new service' projects for BT Payphones and for the PC2000 futures project.

He is a fellow of the Royal Society of Arts, Commerce and Manufacturers.